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Design and Simulation of a Silicon Mach–Zehnder Interferometer

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Abstract This report presents the design and simulation of a silicon photonics Mach–Zehnder Interferometer (MZI) using strip waveguides. In this waveguide eigen mode analysis is done using Lumerical software from which the effective index and group index of the waveguide is extracted and plotted as a function of wavelength. Then a compact model is developed from these data. Then the interferometer transfer function designed using this waveguide is derived analytically and verified using Lumerical INTERCONNECT simulations. Following this various interferometer with varying path difference and TE, TM polarisation are analyzed with their free spectral range (FSR). After this, it is compared with experimental data from the fabricated chip containing the MZI devices, following with their corner analysis.

1 Introduction

brief motivation for silicon photonics. Include references, e.g., [1]. objective of your course project.

Silicon photonics domain has implemented the integration of optical components using CMOS-compatible fabrication processes on large-scale, providing high refractive index contrast, compact footprint, and low-cost manufacturing [[2]] , [[3]] . Among all the silicon photonic integrated circuits, Mach–Zehnder Interferometer (MZI) is widely used for optical modulation, filtering, switching, and sensing applications[[4]] [[5]] .

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In this project we have designed, simulated a MZI, in this we also developed an waveguide compact model for the strip waveguide, and extracted the waveguide effective index and group index from measured transmission spectra. The device is modeled using Lumerical MODE Solutions and INTERCONNECT, The fabrication layout is designed in Klayout software which generates the devices (different MZI's with variable path difference and polarisation i.e, TE TM). The measurement analysis for these devices such as corner analysis is done using the MATLAB.

2 Theory

In its simplest way, a balanced MZI can be model as a 2-port device, formed by a beam-splitter and a beam-combiner, connected by waveguides with identical geometries and material properties. With this configuration, for any wavelength, the signals of the two branches are constructively combined, in the MZI output. Figure 2 illustrates this situation.

An MZI is a 2-port device, that consist of mainly two component y-branch splitter and combiner connected by strip waveguide (in our design) with same material properties. In y-branch splitter the input intensity, I_i, with electric field, $E_i$splits equally into two branches and output of each branch has intensity of equal magnitudes $I_1 = I_2 = I_i/2$, and electric field, $E_1 = E_2 = E_i/\sqrt{2}$ Whereas for combiner is vice-versa, for symmetric case the electric field at the combiner port $E_{o1} = E_1/\sqrt{2}$ or $E_{o2} = E_2/\sqrt{2}$. With these two inputs, the output field will be a vector summation of

the input fields, divided by $\sqrt{2}$, that is

$$E_o = \frac{E_1 + E_2}{\sqrt{2}}.$$

In terms of electric field, lets consider a plane wave, the electric field is represented as $E = E_0 e^{i(\omega t - \beta z)}$, where $\beta = \frac{2\pi n}{\lambda}$, is the propagation constant , n is the index of refraction. This field is modified for the lossless case,

$$E_{o1} = E_1 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1} \tag{1}$$

$$E_{o2} = E_2 e^{-i\beta L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2} \tag{2}$$

Hence the output of the y-branch is

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}\left(e^{-i\beta L_1} + e^{-i\beta L_2}\right) \tag{3}$$

And the intensity is

$$I_o = \frac{I_i}{4}\left|e^{-i\beta L_1} + e^{-i\beta L_2}\right|^2 = \frac{I_i}{2}\left[1 + \cos(\beta \Delta L)\right] \tag{4}$$

2.1 Waveguide Dispersion Model

The propagation constant of a guided mode is given by

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}, \tag{5}$$

where n_{eff} is the effective index of the waveguide mode.

The group index is defined as

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}(\lambda)}{d\lambda}. \tag{6}$$

2.2 Mach–Zehnder Interferometer Transfer Function

In case of unbalanced MZI, it introduces a path length difference ΔL between its two arms. It's the normalized transmission is given as (with assumption of equal power splitting and negligible loss),

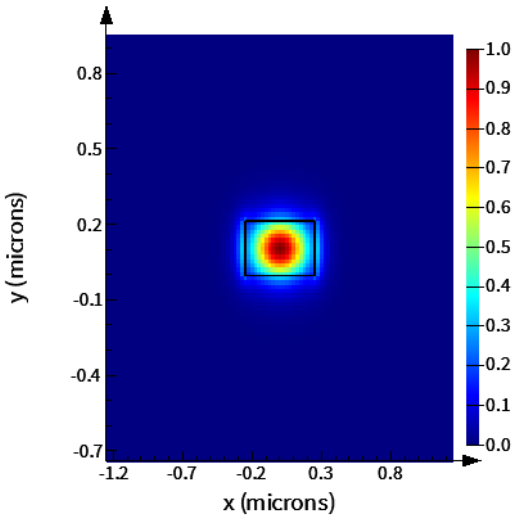
$$T(\lambda) = \frac{1}{2}\left[1 + \cos\left(\beta(\lambda)\Delta L\right)\right]. \tag{7}$$

3 Modelling and Simulation

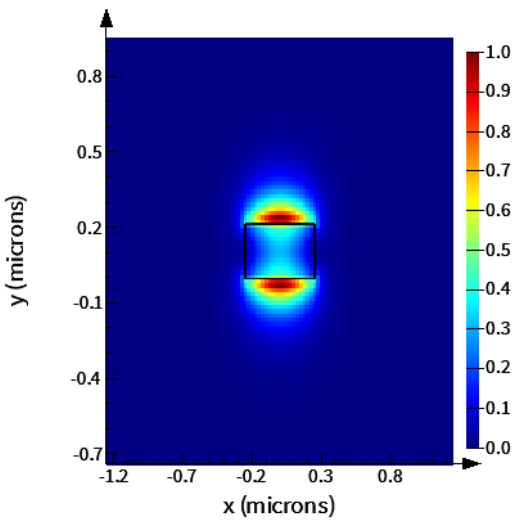
3.1 Waveguide Geometry and Mode Profile

The MZI is designed on silicon on insulator platform using a strip waveguide geometry of cross section 450 nm × 220 nm. For eigen mode analysis the fundamental quasi-TE polarisation is stimulated due to string optical mode confinement. In addition to this we also investigated the TM polarisation for same strip waveguide with same waveguide cross section and same material configuration.

Figure 1 shows the simulated electric field intensity profile of the fundamental TE mode and TM mode obtained using Lumerical MODE Solutions.



(a) Fundamental quasi-TE mode

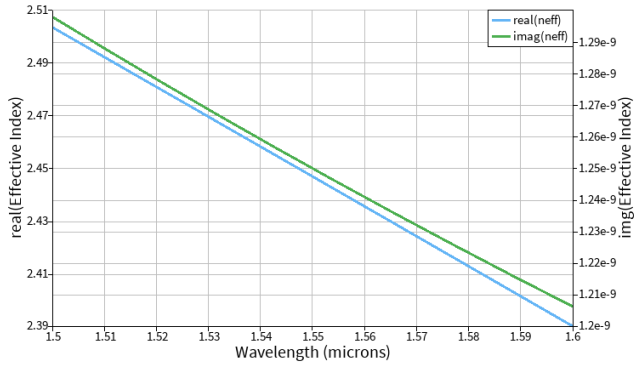


(b) Fundamental quasi-TM mode

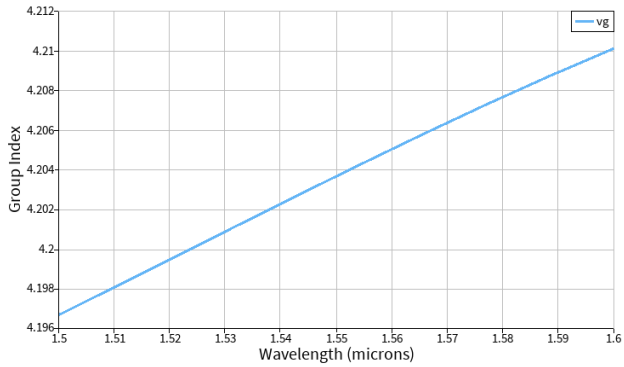
Fig. 1: Simulated electric field intensity profiles of the fundamental modes for the strip waveguide.

3.2 Effective and Group Index

The effective index is extracted over the wavelength range of interest. The group index is computed numerically using the dispersion relation.



(a)



(b)

Fig. 2: (a) Effective index and (b) group index versus wavelength for the TE mode.

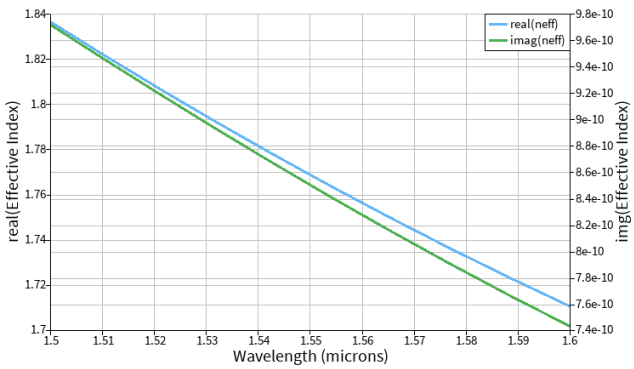
3.3 Compact Waveguide Model

The effective index is approximated using a second-order polynomial expansion around $\lambda_0 = 1.55\ \mu$m:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2. \tag{8}$$

The fitted coefficients are obtained from numerical simulation and used in system-level modeling.
Using Lumerical script editor, the compact model of the waveguide can be calculated, which gives the equation,

$$n_{\text{eff}}(\lambda) = 2.44682 - 1.13339(\lambda - \lambda_0) - 0.0439366(\lambda - \lambda_0)^2. \tag{9}$$



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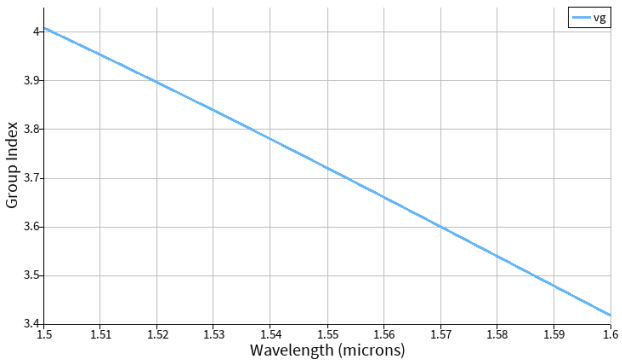


(b)

Fig. 3: (a) Effective index and (b) group index versus wavelength for the TM mode.

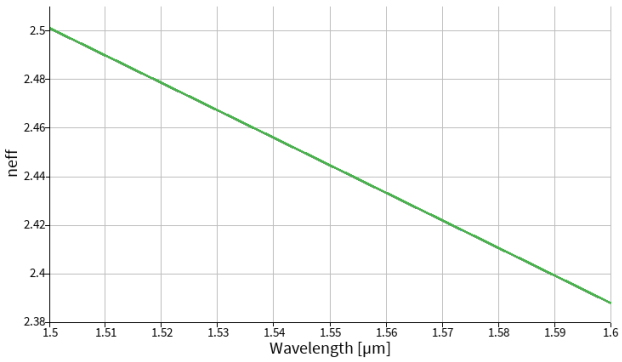


Fig. 5: Compact model of the TE mode, in terms of the effective index. Markers indicate MODE data, and solid lines correspond to a Taylor-series approximation. $n_{\text{eff}}(\lambda) = 2.44682 - 1.13339(\lambda - \lambda_0) - 0.0439366(\lambda - \lambda_0)^2$.

3.4 MZI Parameter Variations

Several MZIs with different path length differences ΔL and polarisation TE and TM are designed to study the dependence of free spectral range on geometry.

Table 1: MZI design parameters and expected performance

ΔL (μm)	Expected FSR (nm)	Cell label
0	0	TE-MZI1
30	17.53	TE-MZI2
60	9.37	TE-MZI3
120	4.80971	TE-MZI4
505	1.0832	TE-MZI5
50	12.38	TM-MZI1
100	6.2	TM-MZI2
150	2.798	TM-MZI3

3.5 Transmission Spectrum

The compact MZI model is designed and simulatted using using Lumerical INTERCONNECT by SiEPIC library components as shown in Figure 4. Figure 6-14 shows the simulated transmission spectrum of various unbalanced MZI (as per table 1) with variable path length difference and polarisation TE TM.

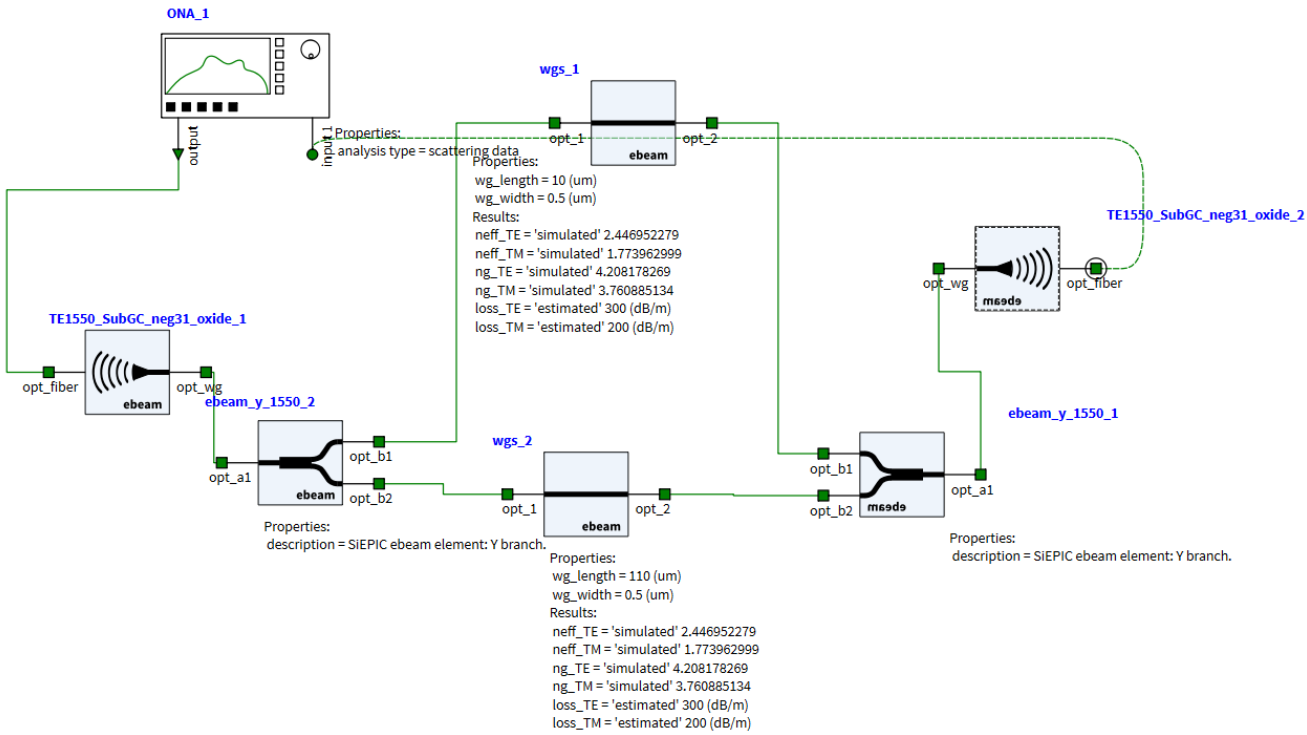
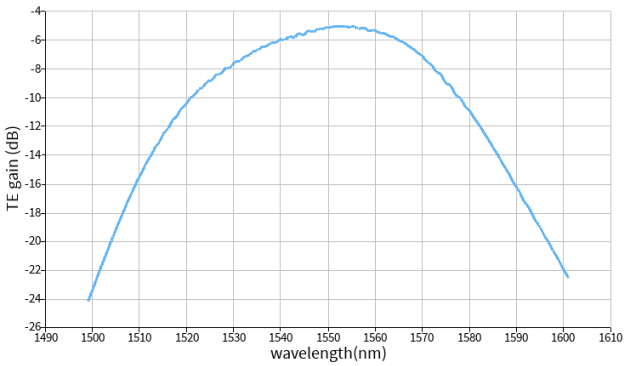
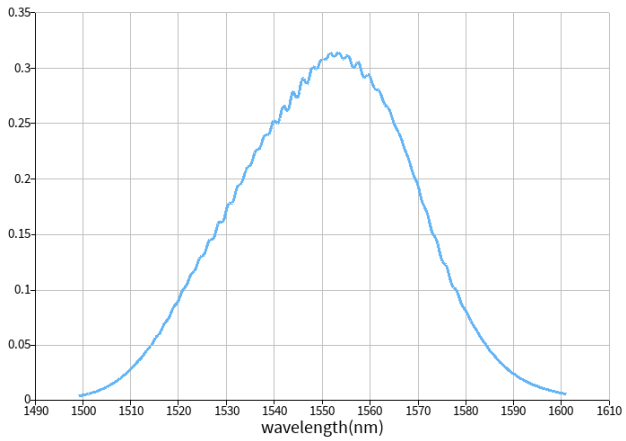


Fig. 4: MZI model implemented in the INTERCONNECT software. ONA: optical network analyzer.

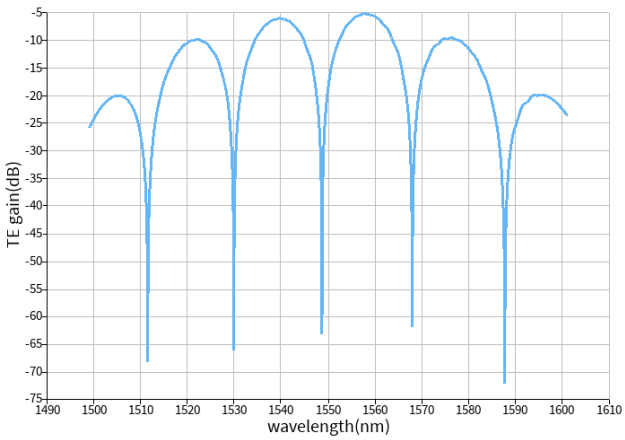


(a)

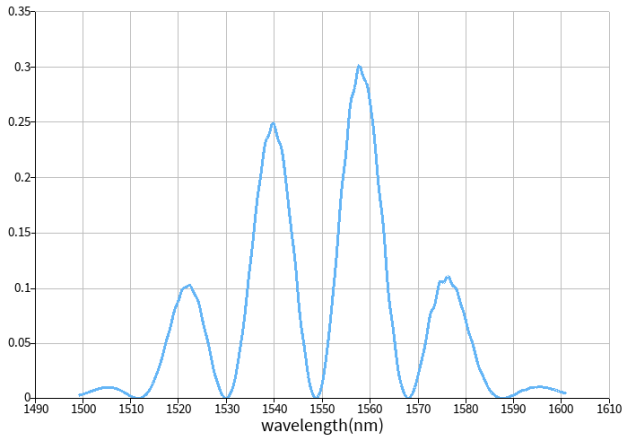


(b)

Fig. 6: (a)Transfer function and (b) Transmission spectrum of MZIs for TE polarisation with length mismatches equal to 0 µm.

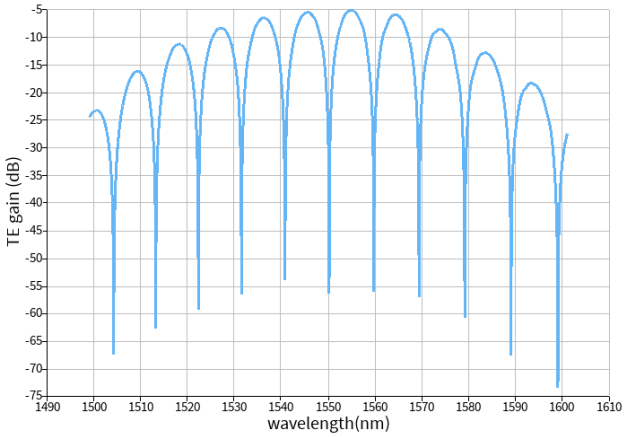


(a)



(b)

Fig. 7: (a)Transfer function and (b) Transmission spectrum of MZIs for TE polarisation with length mismatches equal to 30 µm.



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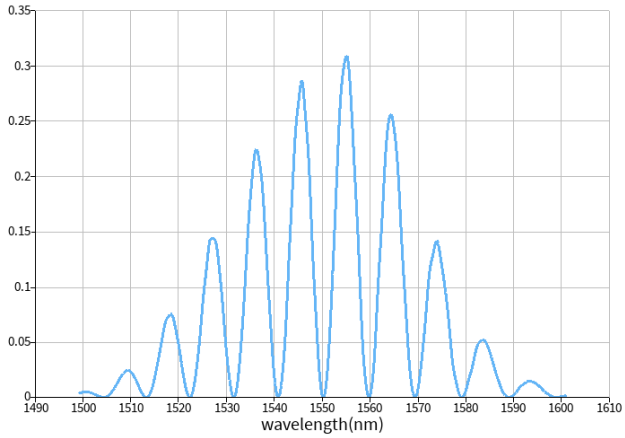
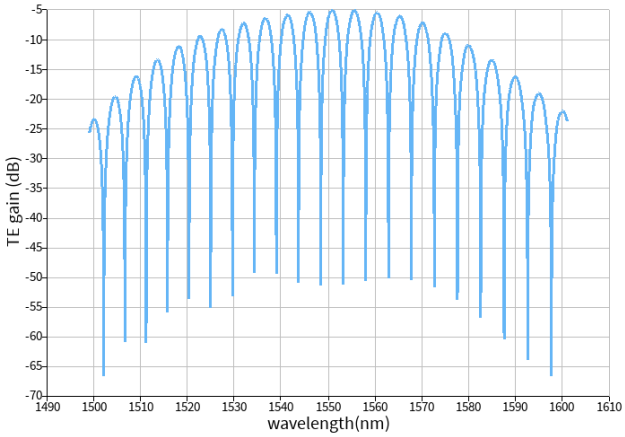


(b)

Fig. 8: (a)Transfer function and (b) Transmission spectrum of MZIs for TE polarisation with length mismatches equal to 60 µm.



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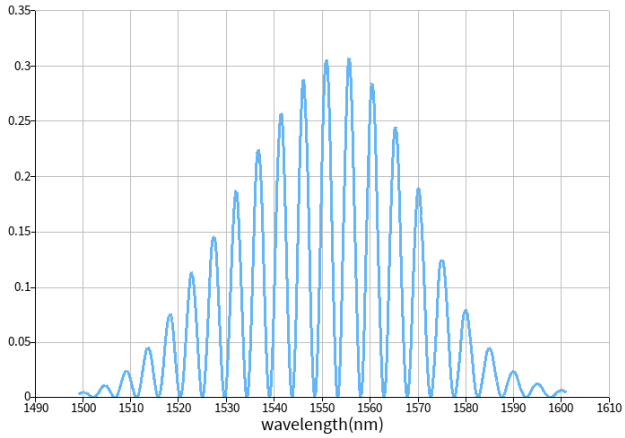


(b)

Fig. 9: (a)Transfer function and (b) Transmission spectrum of MZIs for TE polarisation with length mismatches equal to 120 µm.

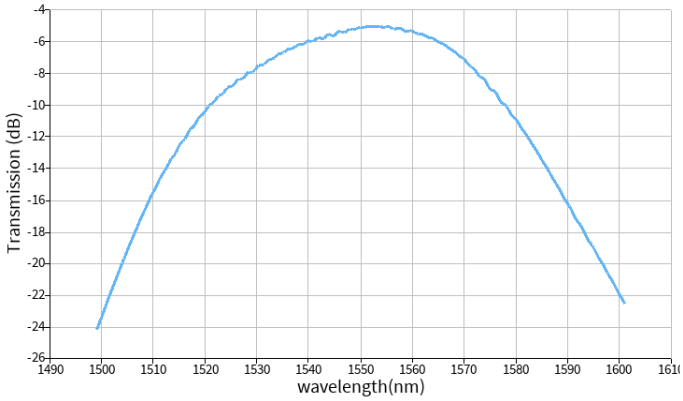
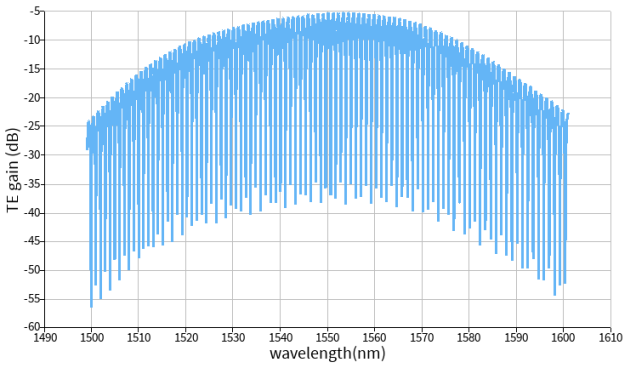
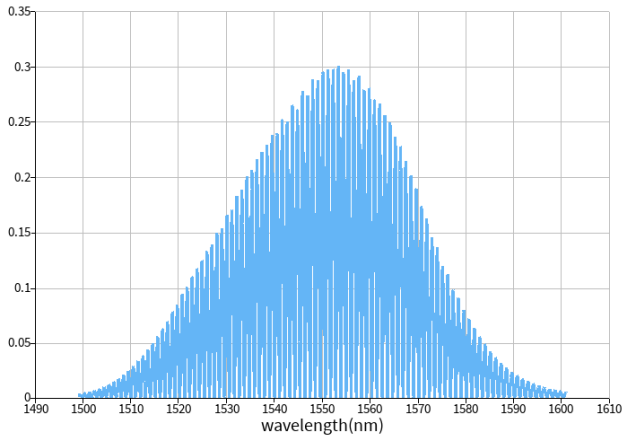


Fig. 10: Simulated transmission spectrum of a Mach–Zehnder Interferometer.

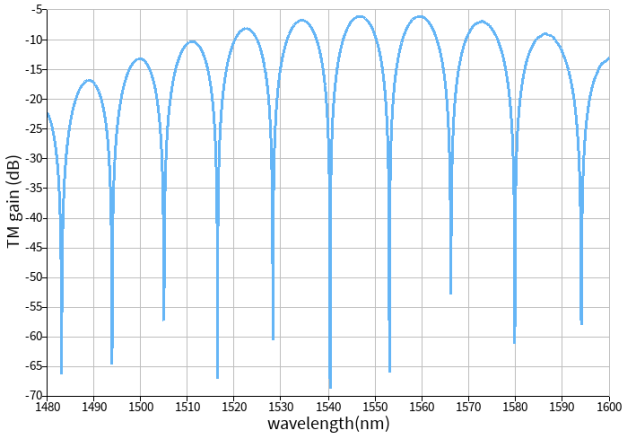


(a)

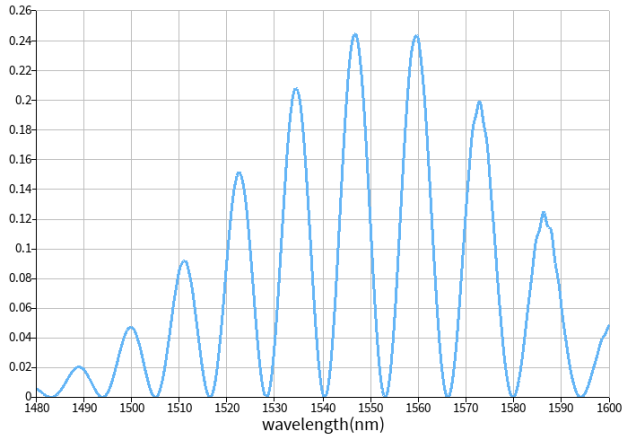


(b)

Fig. 11: (a)Transfer function and (b) Transmission spectrum of MZIs for TE polarisation with length mismatches equal to 505 µm.

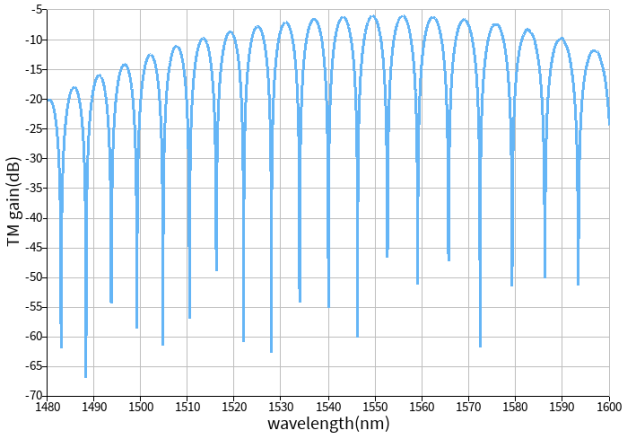


(a)

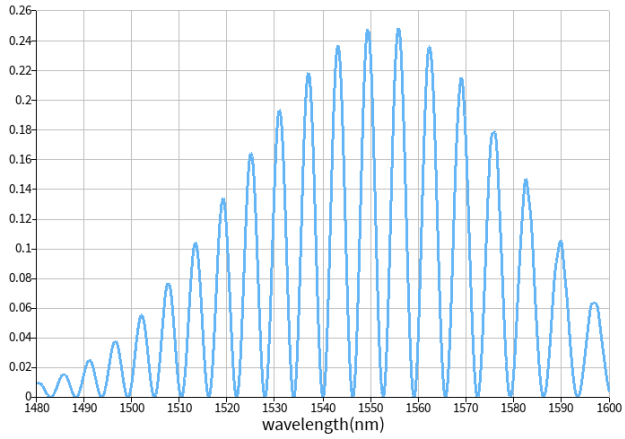


(b)

Fig. 12: (a)Transfer function and (b) Transmission spectrum of MZIs for TM polarisation with length mismatches equal to 50 µm.

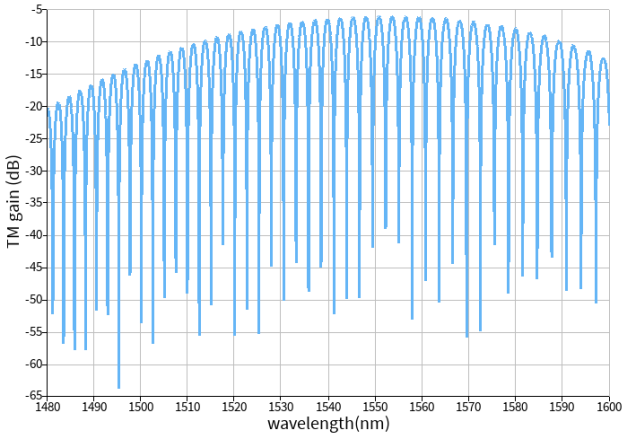


(a)

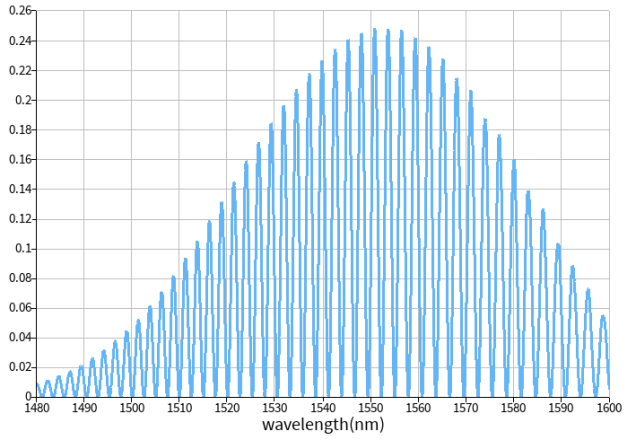


(b)

Fig. 13: (a)Transfer function and (b) Transmission spectrum of MZIs for TM polarisation with length mismatches equal to 100 µm.



(a)



(b)

Fig. 14: (a)Transfer function and (b) Transmission spectrum of MZIs for TM polarisation with length mismatches equal to 150 µm.

4 Layout

The physical layout of the Mach–Zehnder Interferometer (MZI) is designed using KLayout in conjunction with the SiEPIC-EBeam-PDK. In this I have designed the 8 MZI (as per table 1) as shown in Figure. 15 with there de-embedding structures working as caliberation test. In MZI, 5 of them named "TE-MZI1","TE-MZI2","TE-MZI3","TE-MZI4","TE-MZI5" are TE polarised with their de-embedding structure named "TE-Cali". The other 3 MZI named "TM-MZI1","TM-MZI2","TM-MZI3" are TM polarised with their de-embedding structure named "TM-Cali". These MZI have variation of path difference listed in Table 1 with their corresponding FSR values.

5 Fabrication

One chip was fabricated in this course.It is designed using klayout and it's .GDS file was sent for fabrication. The devices (MZI) were fabricated using 100 keV Electron Beam Lithography as described below.

5.1 Washington Nanofabrication Facility (WNF) silicon photonics process:

The devices were fabricated using 100 keV Electron Beam Lithography [[6]]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 µm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 µm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 µC/cm2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil.

5.2 Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (`http://www.appliednt.com/nanosoi`; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

6 Experimental Data

To characterize the devices, a custom-built automated test setup [[7]] with automated control software written in Python was used (`http://siepic.ubc.ca/probestation`). An Agilent 81600B tunable laser was

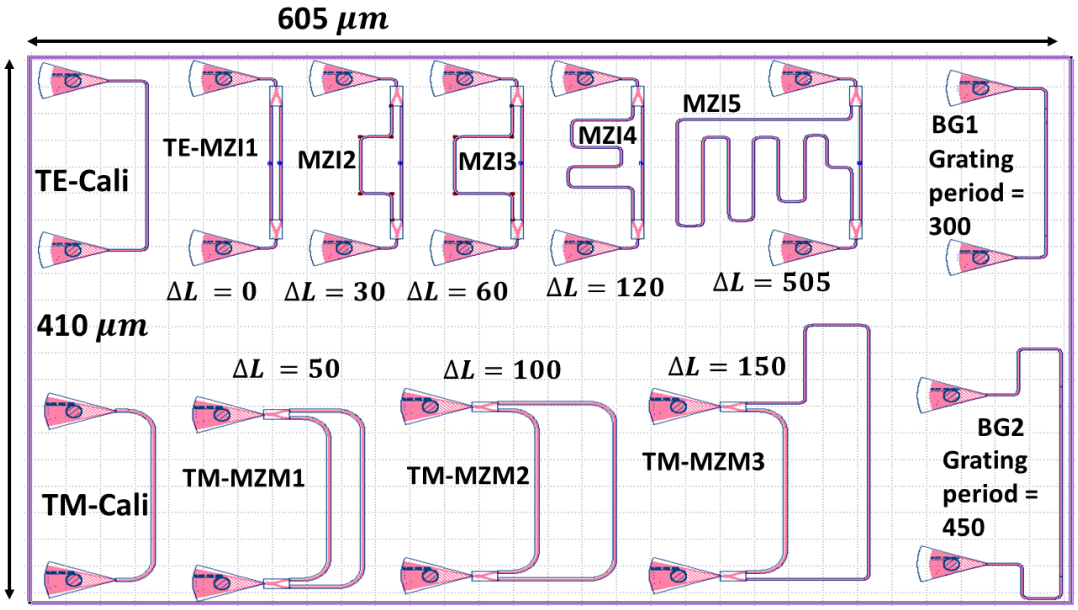


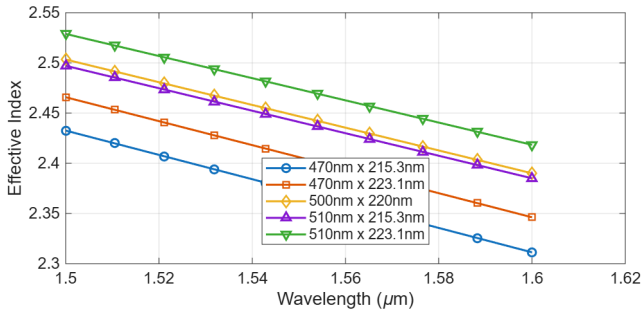
Fig. 15: Final version of the designed devices.

used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [[8]]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [www.plcconnections.com].

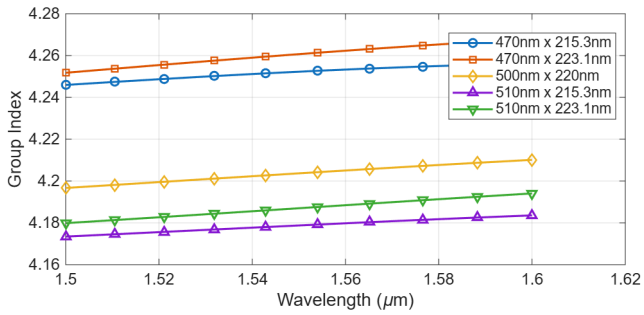
Plots of experimental data. The following figure was generated using a built-in Python interpreter!

7 Manufacturing variability

In terms of fabrication challenges, there is one factor that should be taken into account is " manufacturing variability" that affects the performance of our designed MZI. We can't expect the fabrication to be accurate there are small deviations in fabrication parameters: waveguide width, height and many mor. These variations can lead to variation in effective refractive index, propagation constant. To evaluate the performance of the device with respect to these imperfection, we perform "corner analysis". In this we do analysis of different fabrication parameters: minimum and maximum waveguide width and height used simulated to consider the worst case scenarios. By this analysis we can estimate how sensitive is our device performance (MZI) against manufacturing variability. We see phase shift mismatch with path length difference due to dimension variations, we also observe the shift in interference fringes. Hence, for the MZI stable performance with fabrication tolerance, the corner analysis is must. The corner analysis for our designed MZI is shown in Figure 16-17. As per our corner analysis, the strip waveguide of cross section 500nm $\times$ 220 nm gives the group index between 4.13 and 4.20 for TE mode, and between 3.06 and 3.47 for the TM mode at 1.55 µm.

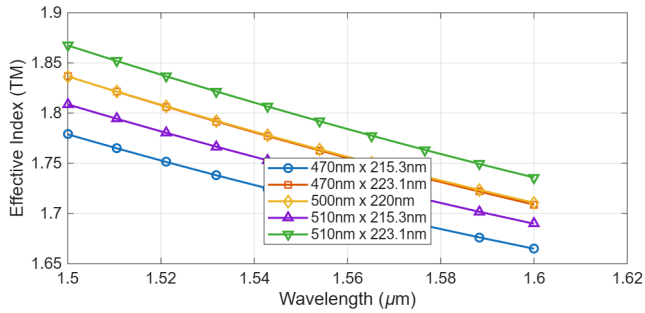


(a)

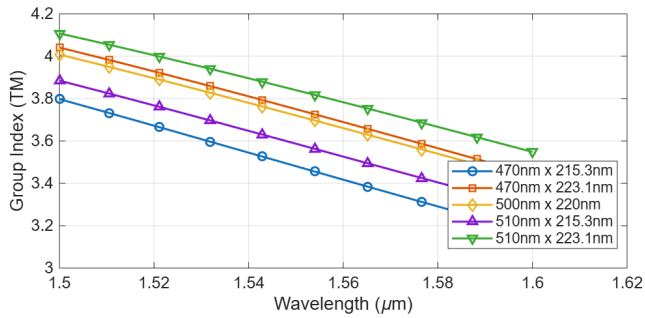


(b)

Fig. 16: (a) Effective index, (b) group index for TE polarisation in strip waveguide with various cross section as per corner analysis.



(a)



(b)

Fig. 17: (a) Effective index, (b) group index for TM polarisation in strip waveguide with various cross section as per corner analysis.

8 Analysis

The free spectral range (FSR) of the MZI is related to the group index by

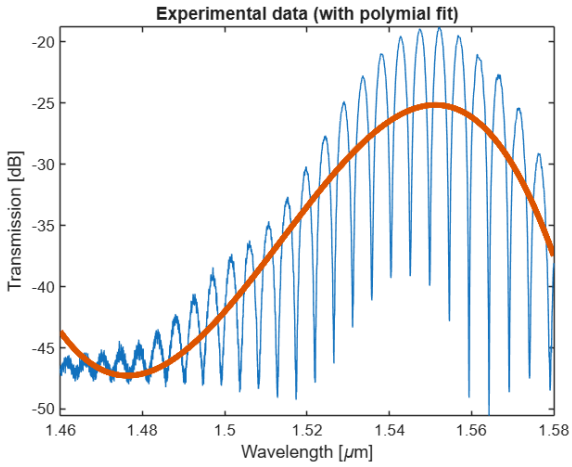
$$\mathrm{FSR} = \frac{\lambda^2}{n_g \Delta L}. \tag{10}$$

Rearranging, the group index can be extracted from measured data as

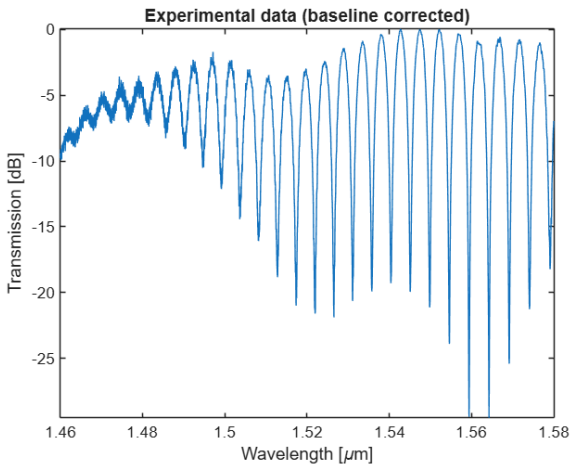
$$n_g = \frac{\lambda^2}{\mathrm{FSR} \cdot \Delta L}. \tag{11}$$

To analyse the performance of our devices, from the experimental results we extracted the waveguide parameters: group index (ng) via curve fitting methods such as autocorrelation fitting to determine the FSR and group index (ng) of our MZI. But first we need ti eliminate the effect of Bragg grating couplers using loop back structures from the transmission spectrum of both MZI (TE and TM). As shown in Figure 15, the “TE-Cali” and “TM-Cali” these are the loop back structures of TE amd Tm polarisation with two grating couplers connected by waveguides. In this report we took the 10 dB variation in loss to counter the noise in the transmission spectrum and this loss is subtracted from the transmission spectrum of our device.

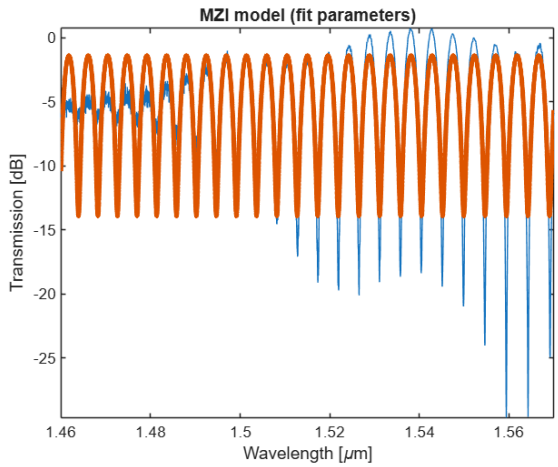
We this we obtain a final calibrated transmission spectrum from which the fabricated waveguide parameters are obtained using curve fitting. The caliberated MZI transmission of our device for TE-polarised 220×500 nm^2 cross section and path length difference of 120 µm is shown in Figure 18 and for TM-polarised 220×500 nm^2 cross section and path length difference of 150 µm is shown in Figure 19. And the extracted group index and FSR for both the MZI is shown in Table 2 and it depicts that the extracted data is well within range as the data from the corner analysis.



(a)

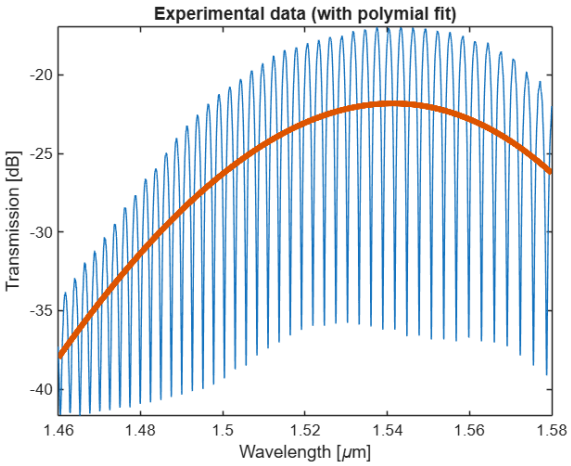


(b)

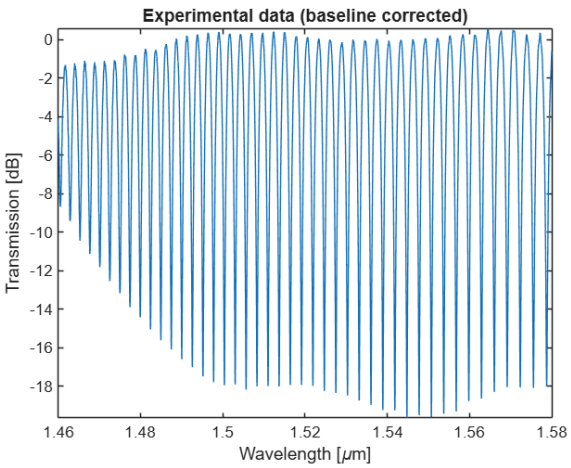


(c)

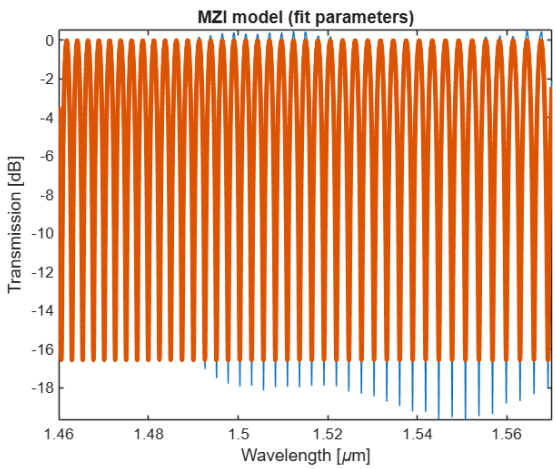
Fig. 18: (a) measurement experimental data with polynomial fit for baseline correction, (b) Experimental (baseline corrected) data and, (c) MZI fit with autocorrelation of the MZI (TE) device with path length difference 120 μm.



(a)



(b)

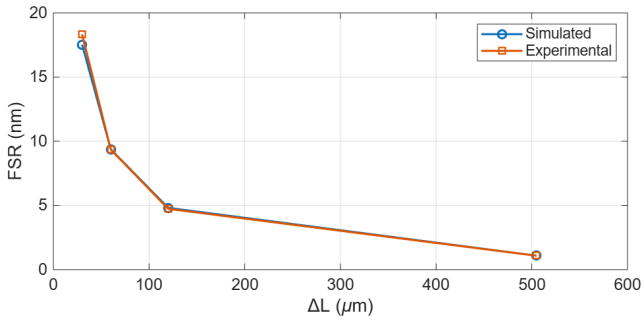


(c)

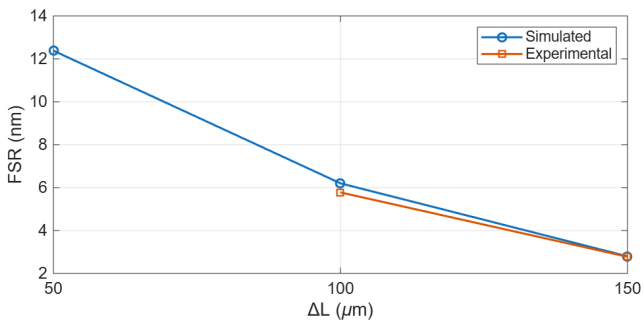
Fig. 19: (a) measurement experimental data with polynomial fit for baseline correction, (b) Experimental (baseline corrected) data and, (c) MZI fit with autocorrelation of the MZI (TM) device with path length difference 150 μm.

Table 2 gives our final results (ng and FSR both found at 1.55 µm) for both simulation and experimental for the different MZI designed by us with variable pathe length length difference and polarisations (TE and TM). The si a larger discrepancies between simulation and experiments for TM mode as compared to TE mode. From this table, the comparisson of FSR for simulated and Fabricated structure vs path length difference is shown in Figure 20.

Cross-section (nm^2)	Mode	ΔL (μm)	FSR (nm) (simulated)	n_g (simulated)	FSR (nm) (experimental)	n_g (experimental)
		0	0		0.46	–
		30	17.53		18.328	4.1744
500 × 220	TE	60	9.37	4.372	9.344	4.0939
		120	4.80971		4.736	4.0386
		505	1.0832		1.0800	4.2083
		50	12.38		device broken	device broken
500 × 220	TM	100	6.2	3.720	5.7760	3.9737
		150	2.798		2.7760	5.5121



(a)



(b)

Fig. 20: FSR versus ΔL. (a) simulated and (b) experimental.

9 Conclusion

This report demonstrated the performance of MZI with TE and TM polarization designed via strip waveguide of cross section 500nm × 220nm. These devices were fabricated using electron beam lithography from University of Washington facility. The MZI were designed using Lumerical software and the final layout for fabrication is designed using the Klayout software. The simulation and experiments results of the MZI were compared using the FSR and group index. We reported an error of 3.72 % in group index for TE mode and less than 6% error for TM MZI's.

10 Acknowledgements

(edit according to your use).

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